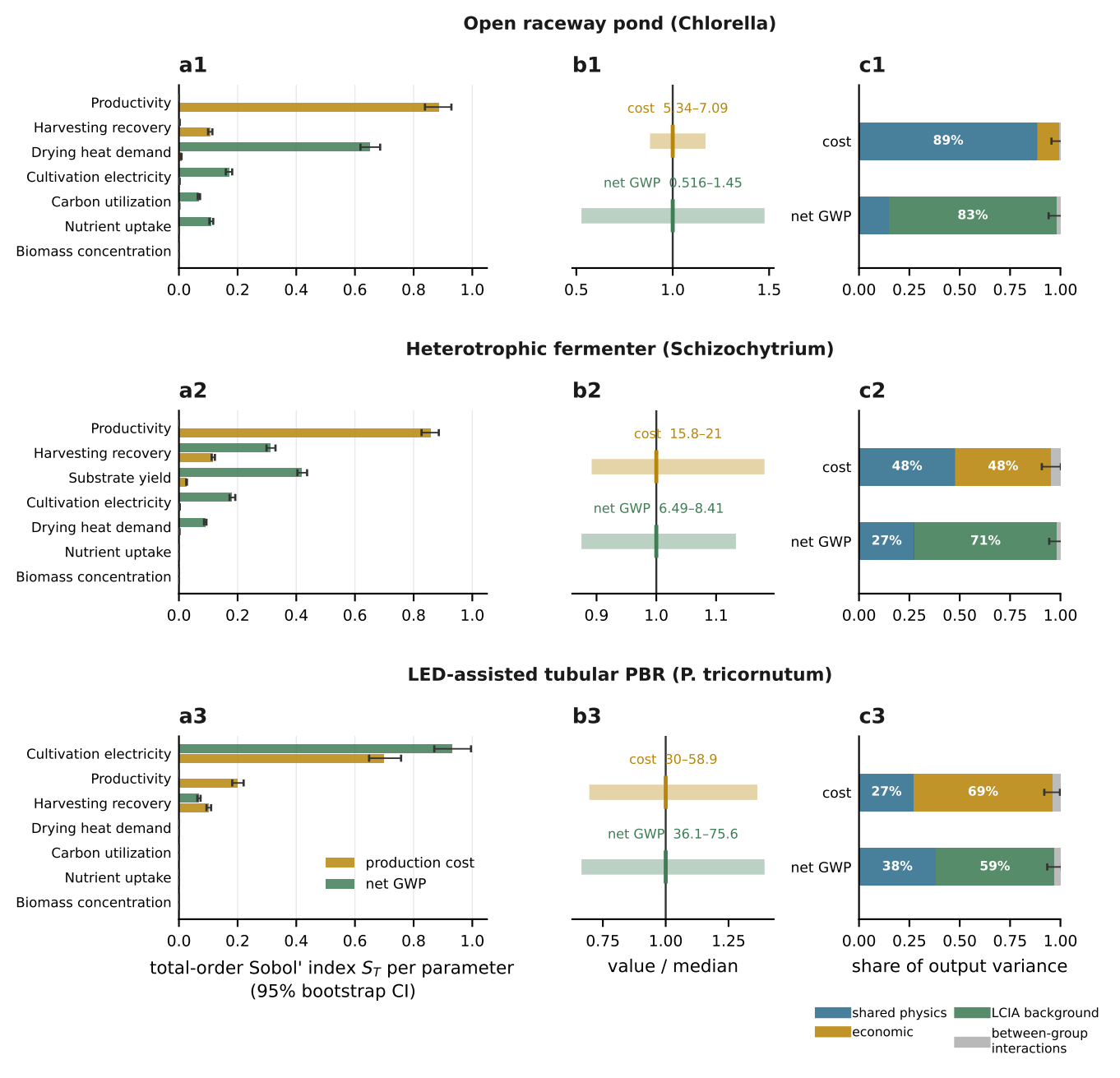
Open raceway pond (Chlorella)
a1
Productivity
Harvesting recovery
Drying heat demand
Cultivation electricity
Carbon utilization
Nutrient uptake
Biomass concentration
0.0
0.2
0.4
0.6
0.8
1.0
b1
cost 5.34–7.09
net GWP 0.516–1.45
0.5
1.0
1.5
c1
cost
net GWP
89%
83%
0.00
0.25
0.50
0.75
1.00
Heterotrophic fermenter (Schizochytrium)
a2
Productivity
Harvesting recovery
Substrate yield
Cultivation electricity
Drying heat demand
Nutrient uptake
Biomass concentration
0.0
0.2
0.4
0.6
0.8
1.0
b2
cost 15.8–21
net GWP 6.49–8.41
0.9
1.0
1.1
c2
cost
net GWP
48%
48%
27%
71%
0.00
0.25
0.50
0.75
1.00
LED-assisted tubular PBR (P. tricornutum)
a3
Cultivation electricity
Productivity
Harvesting recovery
Drying heat demand
Carbon utilization
Nutrient uptake
Biomass concentration
production cost
net GWP
0.0
0.2
0.4
0.6
0.8
1.0
total-order Sobol' index $S_T$ per parameter
(95% bootstrap CI)
b3
cost 30–58.9
net GWP 36.1–75.6
0.75
1.00
1.25
value / median
c3
cost
net GWP
27%
69%
38%
59%
0.00
0.25
0.50
0.75
1.00
share of output variance
shared physics
LCIA background
economic
between-group interactions